

Execute um modelo de IA Generativa localmente

usando Python e Ollama



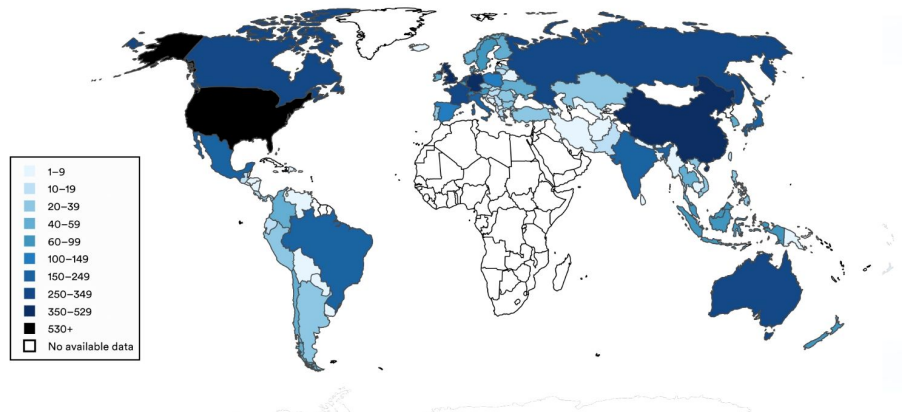
Alerta de tópico sensível

O problema

1. dados usados com permissão?
2. quem gera dados tem remuneração?
3. a privacidade na rede é preservada?
4. o que aconteceu com *whistleblower*?
5. é seguro para a saúde mental?
6. ajuda no desenvolvimento intelectual?
7. faz bem para o meio ambiente?
8. faz sentido substituir pessoas?

Global distribution of data centers, 2025

Source: Cloudscene, 2025 | Chart: 2026 AI Index report



<https://hai.stanford.edu/ai-index/2026-ai-index-report>

A hand is holding a silver metal padlock that is open. The padlock has a keyhole and four screws on its face. In the background, a laptop keyboard is visible, slightly out of focus. The text 'A solução' is overlaid on the image in white.

A solução

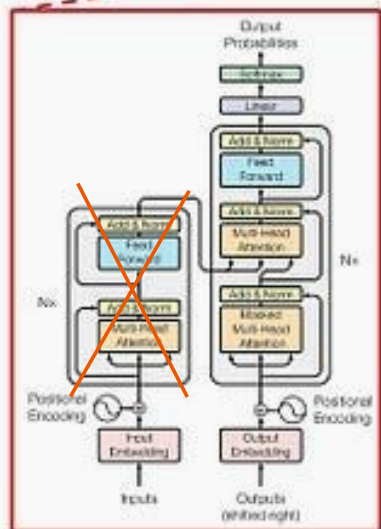
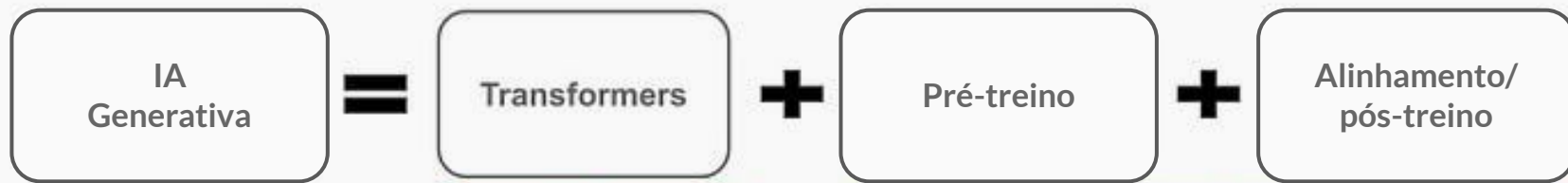
Evitar o uso
caso não tenha jeito,
pelo menos proteger a
privacidade e um pouco do
seu mental executando
modelos de IA Generativa
localmente

A hand is holding a silver metal padlock that is open. The padlock is being held over a laptop keyboard, which is visible in the background. The text 'A solução' is overlaid on the image.

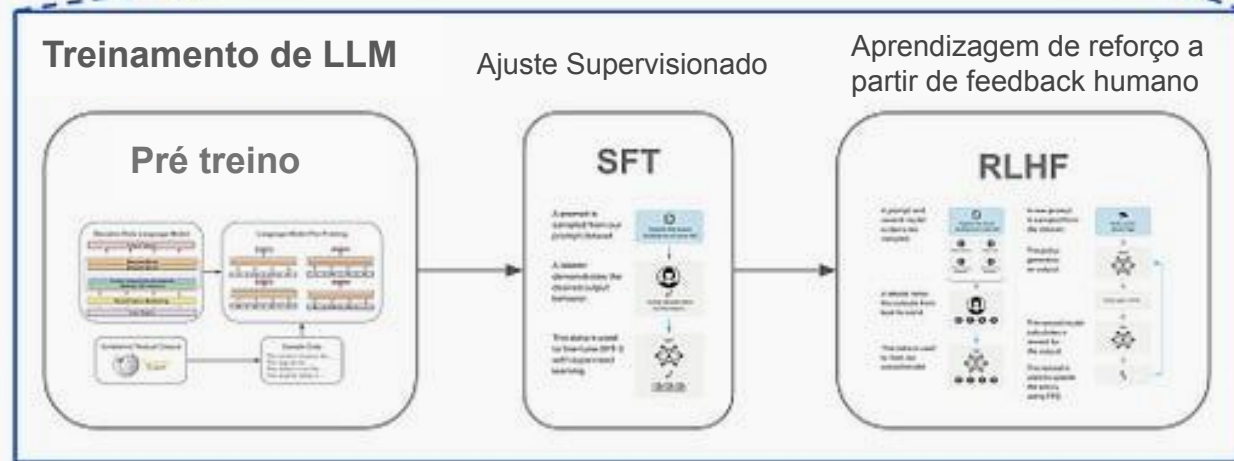
A solução

Neste caso,
seria ideal poder treinar
modelos localmente,
só que para isso é
necessário GPU da NVIDIA
ou chips MAC
M1/M2/M3/M4
e muita memória disponível

Tipos de treinamentos



Apenas decodificador
(decoder only)



feito com transformadores

Supervised Fine Tuning

Reinforcement Learning
from Human Feedback

Como treinar “local”

Unsloth

Notebooks para usar com Colab:

<https://unsloth.ai/docs/get-started/unsloth-notebooks>

Exemplos no TRL do HuggingFace:

<https://huggingface.co/docs/trl/index>



Unsloth Requirements

Here are Unsloth's requirements including system and GPU VRAM requirements.

Unsloth can be used in two ways: through [Unsloth Studio](#), the web UI, or through [Unsloth Core](#), the original code-based version. Each has different requirements.

Unsloth Studio Requirements

- **Mac:** Training, MLX and GGUF inference are ALL supported.
- **CPU:** Unsloth still works without a GPU, for Chat + Data Recipes.
- **Training:** Works on **NVIDIA**, **AMD**, **Intel** GPUs and **Mac** devices

Windows

Unsloth Studio works directly on Windows without WSL. To train models, make sure your system satisfies these requirements:

Requirements

- Windows 10 or Windows 11 (64-bit)
- **NVIDIA GPU with drivers installed**
- **App Installer** (includes `winget`): [here](#) ➤
- **Git:** `winget install --id Git.Git -e --source winget`

só não é pré-requisito para mac



Como treinar “local”

Unsloth

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Taxonomy

Below is the current list of TRL trainers, organized by method type (⚡ = vLLM support; 🛠 = experimental).

Online methods

- [GRPOTrainer](#) ⚡
- [RL00Trainer](#) ⚡
- [OnlineDPOTrainer](#) 🛠 ⚡
- [NashMDTrainer](#) 🛠 ⚡
- [PPOTrainer](#) 🛠
- [XPOTrainer](#) 🛠 ⚡

Reward modeling

- [RewardTrainer](#)
- [PRMTrainer](#) 🛠

Offline methods

- [SFTTrainer](#)
- [DPOTrainer](#)
- [KTOTrainer](#)
- [BCOTrainer](#) 🛠
- [CPOTrainer](#) 🛠
- [ORPOTrainer](#) 🛠

Knowledge distillation

- [DistillationTrainer](#) ⚡
- [GKDTrainer](#) 🛠
- [MiniLLMTrainer](#) 🛠

You can also explore TRL-related models, datasets, and demos in the [TRL Hugging Face organization](#).



Files

main

Go to file

> .github

examples

README.md

async-chat.py

async-structured-outputs.py

async-structured-outputs.py

async-tools.py

chat-logprobs.py

chat-stream.py

chat-with-history.py

chat.py

create.py

embed.py

fill-in-middle.py

generate-image.py

Ollama



Code

Blame

35 lines (32 loc) · 1.21 KB

```
1 from ollama import chat
2
3 messages = [
4     {
5         'role': 'user',
6         'content': 'Why is the sky blue?',
7     },
8     {
9         'role': 'assistant',
10        'content': "The sky is blue because of the way the Earth's atmosphere scatters sunlight.",
11    },
12    {
13        'role': 'user',
14        'content': 'What is the weather in Tokyo?',
15    },
16    {
17        'role': 'assistant',
18        'content': """"The weather in Tokyo is typically warm and humid during the summer months, with tem
19        rarely dropping below freezing. The city is known for its high-tech and vibrant culture, with man
20    },
21 ]
22
23 while True:
24     user_input = input('Chat with history: ')
25     response = chat(
26         'gemma3',
27         messages=[*messages, {'role': 'user', 'content': user_input}],
```

Passo a passo



1. Instalar o Ollama

Acesse

<https://ollama.com/download>
e escolha o seu sistema
operacional

```
ollama --version  
ollama version  
is 0.32.15
```

2. Verificar versão do Ollama

No terminal (Mac/Linux) ou no powershell (Windows), digite `ollama --version` para verificar qual versão foi instalada

```
ollama run smollm  
pulling manifest
```

3. Baixar e rodar o modelo

Digite `ollama run smollm` e baixe o modelo. Sempre veja as suas configurações em <https://ollama.com/search> para ver se tem espaço para o modelo

```
>>>  
>>>  
>>>  
>>>  
>>> Send a message  
(/? for help)
```

4. Usar o modelo

Pode realizar as perguntas livremente. O Ollama salva um arquivo de histórico chamado *history* com todas as interações

1. Instalar o Ollama

```
cassia@galadriel:~$ curl -fsSL https://ollama.com/install.sh | sh
>>> Installing ollama to /usr/local
>>> Downloading ollama-linux-amd64.tar.zst
##### 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/systemd/system/ollama.service.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.
```

2. Verificar versão do Ollama

```
cassia@galadriel:~$ ollama --version  
ollama version is 0.32.15  
cassia@galadriel:~$
```

3. Baixar e rodar o modelo

```
cassia@galadriel:~$ ollama run smollm
```

```
pulling manifest
```

```
pulling 6cafb858555d: 100% ██████████ 990 MB/990 MB 48 MB/s 0s
```

```
pulling 1c3c04e4a658: 100% ██████████ 485 B
```

```
verifying sha256 digest
```

```
writing manifest
```

```
success
```

```
>>>
```

4. Usar o modelo

```
cassia@galadriel:~$ ollama run smollm
>>> hello
Hello! How can I help you today?

>>> are there negative consequences of generative ai use?
Yes, there are several negative consequences of generative AI use,
including job displacement, bias in decision-making, and the potential for
misuse.

>>> Send a message (/? for help)
```

Prática no Colab

1. Login na conta google
2. Abrir o google colab
3. Criar um notebook
4. Seguir o passo a passo

materiais:

https://github.com/cassiasamp/each_ssi_2026

Muito obrigada

Contatos

 @cassiasamp

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